Lecture 1

Complex numbers; vector spaces over $\mathbb{C}$; inner products, norms, orthonormal bases

Instructions. Each question has exactly one correct option. Notation as in the lecture: vectors are bold; $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_j \overline{u_j}\, v_j$; $\mathbf{e}_1 = (1,0)^T$, $\mathbf{e}_2 = (0,1)^T$; $\mathbf{h}_\pm = \frac{1}{\sqrt{2}}(1, \pm 1)^T$; $H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Part A: Complex numbers

Q1. Let $z = \dfrac{1+i}{1-i}$. Then z^{2026} equals

(A) 1 (B) i
(C) -1 (D) $-i$

Q2. If $|z| = 2$, then $\dfrac{1}{z}$ equals

(A) $\overline{z}/4$ (B) $\overline{z}/2$
(C) $z/4$ (D) $-z/4$

Q3. $\dfrac{2+i}{1-2i}$ equals

(A) $\frac{4+5i}{5}$ (B) $-i$
(C) 1 (D) i

Q4. $\left(1 + i\sqrt{3}\right)^6$ equals

(A) -64 (B) 64
(C) $64i$ (D) 32

Q5. A polar form of $-\sqrt{3} - i$ is

(A) $2e^{i\pi/6}$ (B) $2e^{5i\pi/6}$
(C) $2e^{7i\pi/6}$ (D) $2e^{-i\pi/6}$

Q6. Which of the following is **not** a solution of $z^3 = -27$?

(A) $\frac{3}{2}\left(-1 + i\sqrt{3}\right)$ (B) $\frac{3}{2}\left(1 + i\sqrt{3}\right)$
(C) -3 (D) $\frac{3}{2}\left(1 - i\sqrt{3}\right)$

Q7. Real numbers α, β satisfy $\left|e^{i\alpha} + e^{i\beta}\right| = 1$. A possible value of $\alpha - \beta$ is

(A) $\pi/3$ (B) $\pi/2$
(C) π (D) $2\pi/3$

Q8. Let $\omega = e^{2\pi i/5}$. Then $\omega + \omega^2 + \omega^3 + \omega^4$ equals

(A) 0 (B) -1
(C) 1 (D) 4

① $z = \dfrac{1+i}{1-i}$ $\qquad z^{2026}$

$$\frac{(1+i)}{(1-i)} \cdot \frac{(1+i)}{(1+i)} = \frac{1+i^2+2i}{1+1} = \frac{2i}{2} = i$$

$$i^{2026} = i^{2024} \cdot i^2$$

$$= (i^4)^{506} \cdot i^2$$

$i^4 = 1$

$$= 1 \cdot i^2 = -1$$

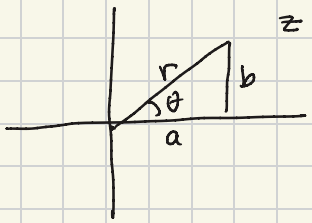
④ $(1+i\sqrt{3})^6$ $\qquad z = 1 + i\sqrt{3}$

$z = re^{i\theta}$

$r = |z| = \sqrt{1+3} = 2$

$\sin\theta = \dfrac{\sqrt{3}}{2}$

$\theta = \pi/3$



$z = a + ib$

$b = \sqrt{3}$

$r = 2$

$\sin\theta = \dfrac{b}{r}$ $\qquad \tan\theta = \dfrac{b}{a}$

$\cos\theta = \dfrac{a}{r}$

$$(1+i\sqrt{3})^6 = \left(2e^{i\pi/3}\right)^6$$

$$= 2^6 \left(e^{i\pi/3}\right)^6$$

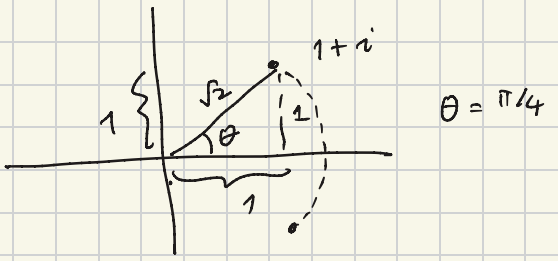
$$= 64\, e^{2i\pi} = 64(\cos 2\pi + i \sin 2\pi)$$

$$= 3 \cdot 64$$

$1+i = \sqrt{2}\, e^{i\pi/4}$

$r = \sqrt{1+1}$

$r = \sqrt{2}$



$\theta = \pi/4$

new $\theta = \frac{\pi}{4} - \frac{\pi}{2} = -\frac{\pi}{4}$

new $r = 2\sqrt{2}$

$\cos(-\phi) = \cos\phi$

$\sin(-\phi) = -\sin\phi$

new complex number $= 2\sqrt{2}\, e^{i(-\pi/4)} = 2\sqrt{2}\left(\cos(-\pi/4) + i\sin(-\pi/4)\right)$

$= 2\sqrt{2}\left(\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right)$

$= \boxed{2 - 2i}$

$-i(1+i)$

$z = re^{i\theta}$

$\frac{1}{z} = \frac{1}{r}\cdot e^{-i\theta}$

$\frac{1}{x} = x^{-1}$

$(e^{i\theta})^{-1} = e^{-i\theta}$

$z \cdot \frac{1}{z} = re^{i\theta}\cdot\frac{1}{r}e^{-i\theta}$

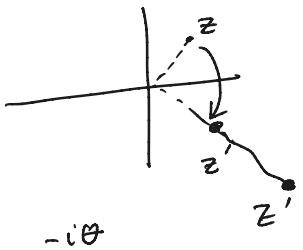
$= e^{i(\theta-\theta)} = ie^{i0}$

$= \cos 0 + i\sin 0 = 1$

Q9. The point $1+i$ is rotated by $90°$ *clockwise* about the origin and its distance from the origin is doubled. The new point is

(A) $2-2i$ (B) $-2+2i$

(C) $2+2i$ (D) $-2-2i$



Q10. For $z = e^{i\theta}$ with θ real, $z - \frac{1}{z}$ equals

(A) $2\sin\theta$ (B) $2\cos\theta$

(C) $2i\sin\theta$ (D) 0

Part B: Vector spaces, inner products

Q11. Which of these is **not** a vector space over $\mathbb{C}$ (with the usual addition and scalar multiplication)?

(A) $\mathbb{C}^3$

(B) the set of unit vectors in $\mathbb{C}^2$

(C) all 2×2 complex matrices

(D) polynomials of degree ≤ 2 with complex coefficients

Q12. Let $\mathbf{u} = \begin{pmatrix}1\\ i\end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix}i\\ -1\end{pmatrix}$. Which statement is true?

(A) $\mathbf{u}, \mathbf{w}$ are linearly independent over $\mathbb{C}$

(B) $\mathbf{u} \perp \mathbf{w}$

(C) $\mathbf{w} = -\mathbf{u}$

(D) $\mathbf{u}, \mathbf{w}$ are linearly dependent over $\mathbb{C}$

Q13. The dimension of $\mathbb{C}^3$ as a vector space over $\mathbb{C}$, and over $\mathbb{R}$, is respectively

(A) 3, 3 (B) 6, 3

(C) 3, 6 (D) 6, 6

Q14. For $\mathbf{u} = \begin{pmatrix}1\\ i\end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix}2\\ i\end{pmatrix}$, the inner product $\langle \mathbf{u}, \mathbf{v}\rangle$ equals

(A) 3 (B) 1

(C) $2+i$ (D) $3i$

Q15. For which $c > 0$ is $c\,(1+i,\ 2,\ i)^T$ a unit vector?

(A) $1/\sqrt{5}$ (B) $1/7$

(C) $1/\sqrt{7}$ (D) $1/\sqrt{3}$

Q16. Suppose $\langle \mathbf{u}, \mathbf{v}\rangle = 2 - i$. Then $\langle i\,\mathbf{u}, 2\mathbf{v}\rangle$ equals

(A) $2+4i$ (B) $-2-4i$

(C) $-2+4i$ (D) $4-2i$

Q17. The coordinates of $\mathbf{v} = (3,1)^T$ in the orthonormal basis $\{\mathbf{h}_+, \mathbf{h}_-\}$ are

(A) (3, 1) (B) (4, 2)

(C) $(\sqrt{2},\ 2\sqrt{2})$ (D) $(2\sqrt{2},\ \sqrt{2})$

$$v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \qquad \frac{\begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}}{\sqrt{2^2+3^2+5^2}}$$

$$\|v\| = \sqrt{|v_1|^2 + |v_2|^2 + \dots + |v_n|^2} = \sqrt{\frac{2}{7}+\frac{4}{7}+\frac{1}{7}} = \sqrt{7}$$

$$\frac{1}{\sqrt{7}}\begin{pmatrix} 1+i \\ 2 \\ i \end{pmatrix} \qquad |v_1| = \sqrt{1+1} = \sqrt{2}$$
$$|v_2| = 2$$
$$|v_3| = 1$$

$$= \begin{pmatrix} \frac{1+i}{\sqrt{7}} \\ \frac{2}{\sqrt{7}} \\ \frac{i}{\sqrt{7}} \end{pmatrix} \qquad \text{e} \quad \frac{\sqrt{7}}{\sqrt{7}} = 1.$$

$\langle u, v \rangle$ $\qquad \langle \alpha u, v \rangle = \bar{\alpha} \langle u, v \rangle$

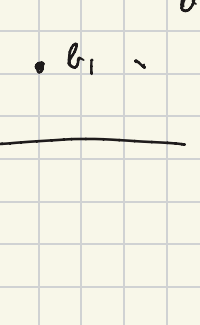
$\langle u, \alpha v \rangle = \alpha \langle u, v \rangle$

$\langle iu, 2v \rangle$

$= -i \langle u, 2v \rangle = -2i \langle u, v \rangle$

Q: Give the coordinates of $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ in the basis $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$

b_1 $\qquad$ b_2



$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

$$= \begin{pmatrix} \alpha + 2\beta \\ \alpha + 5\beta \end{pmatrix}$$

$\alpha + 2\beta = 3$

$\alpha + 5\beta = 1$

$3\beta = -2 \qquad \beta = -2/3$

$\alpha = 3 + 4/3 = \frac{13}{3}$

$\begin{pmatrix} \frac{13}{3} \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -\frac{2}{3} \\ 1 \end{pmatrix}$

What happens the basis is orthonormal? b_1, b_2

$$\langle b_1, b_1 \rangle = 1, \quad \langle b_2, b_2 \rangle = 1, \quad \langle b_1, b_2 \rangle = 0$$

$$v = \alpha b_1 + \beta b_2$$

$$\langle b_1, v \rangle = \langle b_1, \alpha b_1 \rangle + \langle b_1, \beta b_2 \rangle$$

$$= \alpha \langle b_1, b_1 \rangle + \beta \langle b_1, b_2 \rangle$$

$$= \alpha$$

$$\langle b_2, v \rangle = \beta$$

Q18. Show that if $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$, where $k \leq n$, are k linearly independent vectors in $\mathbb{C}^n$, then there is a basis of $\mathbb{C}^n$ which includes these vectors.

Q19. Let $W = \{(z_1, z_2, z_3)^T \in \mathbb{C}^3 : z_1 + iz_2 - z_3 = 0\}$.

(a) Show that W is a subspace of $\mathbb{C}^3$. [3]

(b) Find a basis of W and state $\dim_{\mathbb{C}} W$. Justify that your vectors are linearly independent. [4]

(c) Is $S = \{(z_1, z_2)^T \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}$ a subspace of $\mathbb{C}^2$? Justify, and say briefly why this matters for quantum states. [3]

Q20. Let $\mathbf{u} = \begin{pmatrix} 1 \\ i \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 1+i \\ i-1 \end{pmatrix}$.

(a) Show that $\mathbf{u}, \mathbf{w}$ are linearly *dependent* over $\mathbb{C}$ by finding $\alpha \in \mathbb{C}$ with $\mathbf{w} = \alpha\mathbf{u}$.

(b) Show that $\mathbf{u}, \mathbf{w}$ are linearly *independent* over $\mathbb{R}$.

* (c) What is the dimension of $\mathbb{C}^2$ as a vector space over $\mathbb{R}$? Write down an explicit real basis.

Q21. Let $\mathbf{u} = (1,\ i,\ 1-i)^T$ and $\mathbf{v} = (i,\ 2,\ 1)^T$ in $\mathbb{C}^3$.

(a) Compute $\langle \mathbf{u}, \mathbf{v} \rangle$, $\langle \mathbf{v}, \mathbf{u} \rangle$, $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$, and verify the Cauchy–Schwarz inequality.

(b) Find the component of $\mathbf{v}$ orthogonal to $\mathbf{u}$, and check that it is orthogonal.

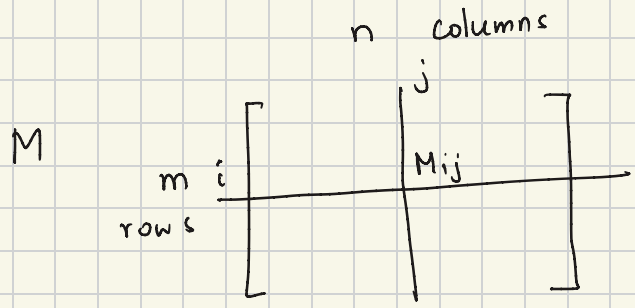
(c) Using only conjugate symmetry and linearity in the second slot, prove that $\langle \alpha\mathbf{u}, \mathbf{v} \rangle = \overline{\alpha}\, \langle \mathbf{u}, \mathbf{v} \rangle$ for all $\alpha \in \mathbb{C}$.

$$u = \frac{1}{2}\begin{pmatrix} 1 \\ i \\ 1-i \end{pmatrix}$$

$$u^\dagger = \frac{1}{2}\begin{pmatrix} 1 & -i & 1+i \end{pmatrix}$$

$$uu^\dagger = \frac{1}{4}\begin{pmatrix} 1 \\ i \\ 1-i \end{pmatrix}\begin{pmatrix} 1 & -i & 1+i \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 1 & -i & 1+i \\ i & 1 & i-1 \\ 1-i & -i-1 & 2 \end{pmatrix}$$

$uu^\dagger$



How do we add and multiply matrices?

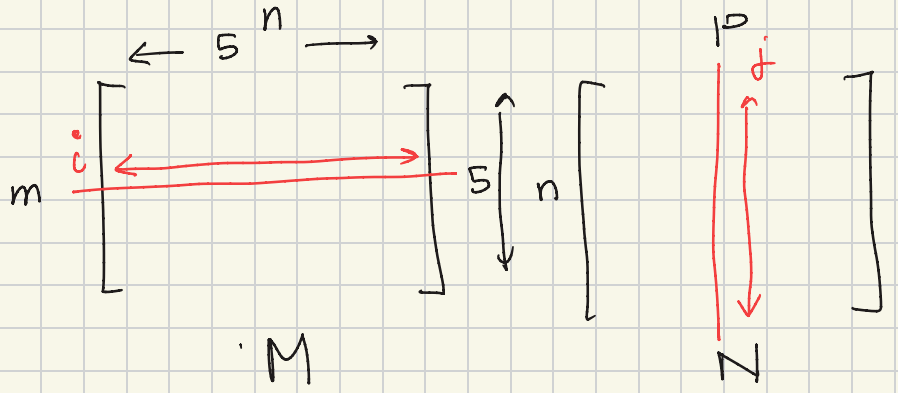
$m \times n$

$$\begin{bmatrix} 4 & 7 & 6 \\ i & 3i & 1+i \end{bmatrix} + \begin{bmatrix} 1 & 0 & i \\ 2 & 3 & -i \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 7 & 6+i \\ 2+i & 3+3i & 1 \end{bmatrix}$$

$M \times N$

$Q =$



$$M_{i1} * N_{1j} + M_{i2} * N_{2j} + \cdots + M_{in} * N_{nj}$$

columns of M = # rows of N

$m \times \underline{n} \quad \underline{n} \times p$

$Q: \quad m \times p$

What will Q_{ij} look like? Consider i^{th} row of M, j^{th} col of N

$Q =$ M $\begin{bmatrix} 2 & 1 & 5 \\ 6 & 0 & 1 \end{bmatrix}$ 2×3 $\quad$ N $\left[\begin{array}{c|c} 1 & 2 \\ 5 & 1 \\ 2 & 3 \end{array}\right]$ 3×2

$M^T = \begin{bmatrix} 2 & 6 \\ 1 & 0 \\ 5 & 1 \end{bmatrix}$ 3×2

$Q : 2\times 2$

$Q_{11} = 2\times 1 + 1\times 5 + 5\times 2$
$= 17$

$Q_{12} = 2\times 2 + 1\times 1 + 5\times 3 = 20$

$Q_{21} = 6\times 1 + 0\times 5 + 1\times 2 = 8$

$Q_{22} = 6\times 2 + 0\times 1 + 1\times 3 = 15$

$$Q = \begin{bmatrix} 17 & 20 \\ 8 & 15 \end{bmatrix}$$

$\langle u, v \rangle$

$\begin{bmatrix} 5 & 1 & 3 \end{bmatrix}_{1\times 3} \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}_{3\times 1}$ $\quad V = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$

$1\times 1 = [\ ]$ $\quad n\times 1$ matrix

$5\times 6 + 1\times 2 + 3\times 1$

$\langle u, v \rangle = \bar{u}_1 v_1 + \bar{u}_2 v_2 + \cdots + \bar{u}_n v_n$

$\begin{bmatrix} \bar{u}_1 & \bar{u}_2 & \cdots & \bar{u}_n \end{bmatrix}_{1\times n} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}_{n\times 1}$ $\quad u^\dagger = \bar{u}^T$

$\langle u, v \rangle = \underset{1\times n}{\bar{u}^T} \cdot \underset{n\times 1}{v} = \underset{1\times n}{u^\dagger} \cdot \underset{n\times 1}{v}$

$$v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \qquad u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$$

$u^T v$ $\qquad$ uv^T

1×1

$$\begin{bmatrix} \\ u \\ \\ \end{bmatrix} \begin{bmatrix} v^T \end{bmatrix} \qquad n \times n$$

$n \times 1$ $\qquad$ $1 \times n$

$$u = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

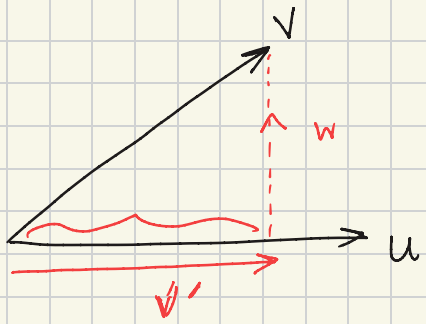
uu^T

$$uu^T = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} (1\ 2\ 3)$$

$$uu^T = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$$

"outer product"

Q:



$v = \alpha u + w$ (double-headed arrow)

such that $\langle u, w \rangle = 0$

Find the component of v along u?

Find the projection of v along u?

Assume that u is a unit vector

along u / perpendicular to u

$v = \alpha u + w$ $\qquad$ $v = \underbrace{\langle u, v \rangle}_{\text{complex}} u + w$

$\langle u, v \rangle = \alpha \langle u, u \rangle + \langle u, w \rangle$

$\alpha = \langle u, v \rangle$

$\langle u, v \rangle = \alpha |u|^2 \qquad \alpha = \frac{\langle u, v \rangle}{|u|^2}$

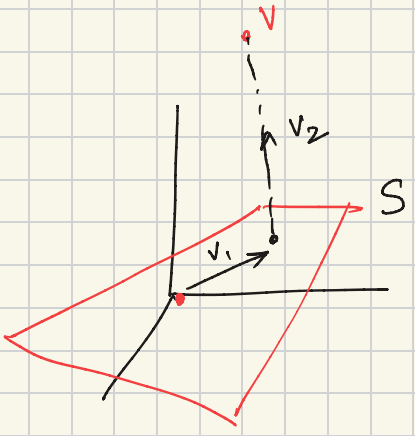
Component of v along u $= (u^\dagger v)\, u$

$= u u^\dagger v$ — "Projection Matrix"

$= (u u^\dagger) v$

$n \times n$ matrix P

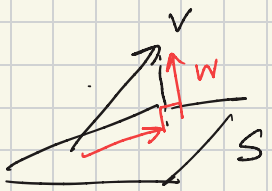
$[\ ][\ \ \ ]$



$v = v_1 + v_2$

v_1 lies in S, $\langle v_1, v_2 \rangle = 0$

Given a subspace S, how do we project a vector v along this subspace?



Idea: Take an orthonormal basis of S

$b_1, b_2, \dots, b_k$ $\quad \langle b_i, b_i \rangle = 1$

$\langle b_i, b_j \rangle = 0$ if $i \neq j$

$$v = \underbrace{\alpha_1 b_1 + \alpha_2 b_2 + \dots + \alpha_k b_k}_{\text{component of } v \text{ along } S} + w$$

w: perpendicular to $b_1, \dots, b_k$

$\langle b_i, v \rangle = \alpha_i$

$\Rightarrow$ component of v in this basis

$$= \langle b_1, v \rangle b_1 + \dots + \langle b_k, v \rangle b_k$$

$$= \underbrace{\left(b_1 b_1^T + b_2 b_2^T + \dots + b_k b_k^T \right)}_{\text{projection matrix}} v$$

$[\ ][\ \ ]$

Example: $\mathbb{R}^3$ plane through the origin and the points

$$\underbrace{\frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix}}_{b_1} \quad \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\\0\end{pmatrix}}_{b_2} \qquad b_1 = \begin{pmatrix}1\\1\\1\end{pmatrix} \quad b_2 = \begin{pmatrix}1\\2\\3\end{pmatrix}$$

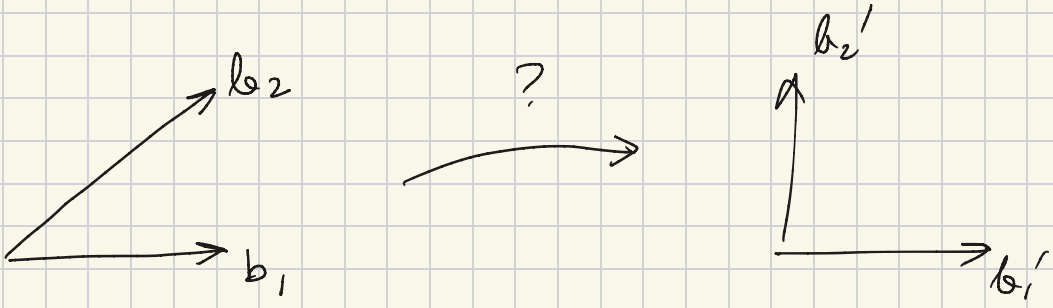
Q: what is the component of $\begin{pmatrix}3\\2\\5\end{pmatrix}$ along this plane? $\perp$ to this plane?

$$b_1 \langle b_1, v \rangle + b_2 \langle b_2, v \rangle = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix} \cdot \frac{10}{\sqrt{3}} + \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\\0\end{pmatrix}\frac{1}{\sqrt{2}}$$

Does every subspace have an orthonormal basis?

How do we find it?

There is a procedure which given a basis converts it into an orthonormal basis.



$\text{span}(b_1, b_2) = \text{span}(b_1', b_2')$ and

b_1', b_2' are orthonormal.

$$b_1' = \frac{b_1}{\|b_1\|}$$

b_2' = component of b_2 perpendicular to b_1

b_3' = component of b_3 perpendicule to b_1', b_2'

$\mathbb{R}^4$

$$b_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \qquad b_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} \qquad v^\dagger = v^*$$

S = plane spanned by b_1, b_2 in $\mathbb{R}^4$

Q: what is the component of $\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ along S?

v

$b_1, b_2 \longrightarrow$ orthonormal basis b_1', b_2'

$$b_1' = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \qquad b_2' = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$$

component of b_2 along b_1'

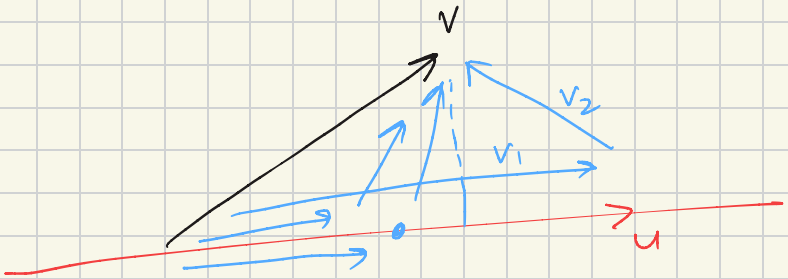
$$= \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \langle b_1', b_2 \rangle$$

$$= \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \cdot 2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

component of $b_2 \perp b_1' = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$

Component of any vector v along S

$$= b_1' \langle b_1', v \rangle + b_2' \langle b_2', v \rangle = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \cdot \frac{1}{2} + \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} \cdot \frac{1}{\sqrt{2}}$$


V
v_2
v_1
u

Answer key

Q	Ans	Explanation
1	C	$z = \frac{(1+i)^2}{(1-i)(1+i)} = \frac{2i}{2} = i$, and $2026 = 4 \cdot 506 + 2$, so $z^{2026} = i^2 = -1$.
2	A	$\frac{1}{z} = \frac{\overline{z}}{\lvert z\rvert^2} = \frac{\overline{z}}{4}$.
3	D	$\frac{(2+i)(1+2i)}{(1-2i)(1+2i)} = \frac{2+5i+2i^2}{5} = \frac{5i}{5} = i$. (A) comes from using $i^2 = +1$.
4	B	$1 + i\sqrt{3} = 2e^{i\pi/3}$, so the sixth power is $64\, e^{2\pi i} = 64$.
5	C	$r = 2$; the point is in the third quadrant, so $\theta = 7\pi/6$. $\tan^{-1}(b/a)$ alone gives $\pi/6$ (A), the wrong quadrant.
6	A	The roots are $3e^{i\pi/3}, -3, 3e^{-i\pi/3}$, i.e. (B), (C), (D). Option (A) is $3e^{2\pi i/3}$, whose cube is $+27$.
7	D	$\lvert e^{i\alpha} + e^{i\beta}\rvert^2 = 2 + 2\cos(\alpha - \beta) = 1$ gives $\cos(\alpha - \beta) = -\frac{1}{2}$.
8	B	The five 5th roots of unity sum to 0, and the omitted one is $\omega^0 = 1$.
9	A	Multiply by $2e^{-i\pi/2} = -2i$: $(1 + i)(-2i) = -2i - 2i^2 = 2 - 2i$.
10	C	$\lvert z\rvert = 1$, so $\frac{1}{z} = \overline{z} = e^{-i\theta}$, and $e^{i\theta} - e^{-i\theta} = 2i \sin\theta$.
11	B	Not closed under addition: $\mathbf{e}_1 + \mathbf{e}_1$ has length 2.
12	D	$i\,\mathbf{u} = (i, i^2)^T = \mathbf{w}$. Also $\langle \mathbf{u}, \mathbf{w}\rangle = i + (-i)(-1) = 2i \neq 0$, so (B) is false.
13	C	Over $\mathbb{R}$ the real and imaginary parts of each entry count separately.
14	A	$\overline{1} \cdot 2 + \overline{i} \cdot i = 2 + 1 = 3$. (B) is the result without the conjugate.
15	C	$\lVert(1 + i, 2, i)^T\rVert^2 = \lvert 1 + i\rvert^2 + 4 + \lvert i\rvert^2 = 2 + 4 + 1 = 7$.
16	B	$\langle i\mathbf{u}, 2\mathbf{v}\rangle = \overline{i} \cdot 2\, \langle \mathbf{u}, \mathbf{v}\rangle = -2i(2 - i) = -4i + 2i^2 = -2 - 4i$. (A) forgets to conjugate i.
17	D	$\langle \mathbf{h}_+, \mathbf{v}\rangle = 4/\sqrt{2} = 2\sqrt{2}$ and $\langle \mathbf{h}_-, \mathbf{v}\rangle = 2/\sqrt{2} = \sqrt{2}$.
18	B	Column j of the matrix is $T\mathbf{e}_j$. (A) is the transpose.
19	A	Circuits are read left to right; matrix products act right to left.
20	D	This is H; its columns are orthonormal. (A): columns not orthogonal; (B): a projector; (C): det = 0, not even invertible.